SLU O607 DVT TEAM 17

Date: 07/20/2026

Group Members

ANUJ KUMAR (ak2305493@gmail.com)

David Kimani (cheged621@gmail.com)

Pheeha Rafahlema (pheeha69@gmail.com)

NWOHIRI INNOCENT (inwohiri@gmail.com)

Srijon Ghosh (srijon.gh82@gmail.com)

Praneel Reddy Kanduri (kpraneelreddy18@gmail.com)

Kamogelo Harry Samotse (kamogeloharry089@gmail.com)

Jian Kalel Marquez (kaleldmarquez@gmail.com)

Sudeep Hebbar (hebbarsudeep@gmail.com)

1. Introduction and Dataset Overview

1.1 Executive Summary

This exploratory analysis of 5,728 cleaned Excelerate opportunity records shows a platform dominated by Internship, Career, and Competition opportunities delivered primarily through Work From Home and Virtual modes, reflecting a strong focus on accessible, employability-driven offerings. Most opportunities are low-cost, with 85.8% including microscholarship support and near-zero median fees overall, though Program and XploreU opportunities break this pattern with the highest fees and shortest durations, and extreme outliers in Fee and Microscholarship required median-based rather than mean-based interpretation. The dataset is well-structured for exploratory work but has two limitations addressed in Section 4: missing location data (25.12%) and the absence of outreach-channel or applicant-level variables.

1.2 Data Preparation / Methodology

1.2.0 Data Preparation and Methodology

The raw Opportunity dataset was cleaned and processed using Python/pandas prior to analysis. Cleaning steps included removing duplicate records, standardizing date formats for created_at and last_date_to_apply, and reducing the original set of variables to the 15 retained for analysis (2 date, 3 numerical, 10 categorical) based on relevance to opportunity-level characteristics. No imputation was performed on missing values; instead, missingness was quantified and reported directly (**Section 1.2.2**) so that affected variables — particularly location — could be interpreted with appropriate caution.

1.2.1 Dataset Summary

The cleaned Opportunity dataset file Opportunity_Data_cleaned.xlsx analyzed in this report consists of 5,728 cleaned opportunity records and 16 retained variables, where each record represents a single opportunity available on the Excelerate platform. The dataset contains information related to opportunity categories, publication dates, application deadlines, duration, fees, currency types, locations, microscholarships, and approval status. It includes 10 opportunity categories, **2 date variables** (`created_at` and `last_date_to_apply`), **3 numerical variables** (`duration`, `fee`, and `microscholarship`), and **10 categorical variables**. No duplicate records remained after the data cleaning process, and the primary unit of analysis is **one opportunity per record.**

1.2.2 Missing Value Analysis

Variables	Missing Values	Percentage (%)
location	1,439	25.12
long_ description	344	6.01
currency_ type	340	5.94
duration_ type	2	0.03
modified_at	2	0.03
code	1	0.02
create_at	1	0.02
last_date_ to_apply	1	0.02
microscholarship	1	0.02
name	1	0.02
short_ description	1	0.02

Table 1: Present the number and percentage of missing values identified across the variables in the cleaned Opportunity dataset

Explanation & Interpretation: The missing values analysis was conducted to assess the completeness of the cleaned Opportunity dataset. The table and accompanying bar chart show that missing data is concentrated in a small number of variables, with Location recording the highest proportion of *missing values at **25.12%**,* followed by *Long Description **(6.01%)*** and *Currency Type* ***(5.94%).*** All remaining variables contain negligible levels of missing values, generally below ***0.1%***. The overall pattern indicates that the dataset is largely complete and suitable for exploratory data analysis, although findings related to location should be interpreted with caution due to the relatively high proportion of missing location information.

2. Exploratory Data Analysis

Distribution of Opportunities by Category

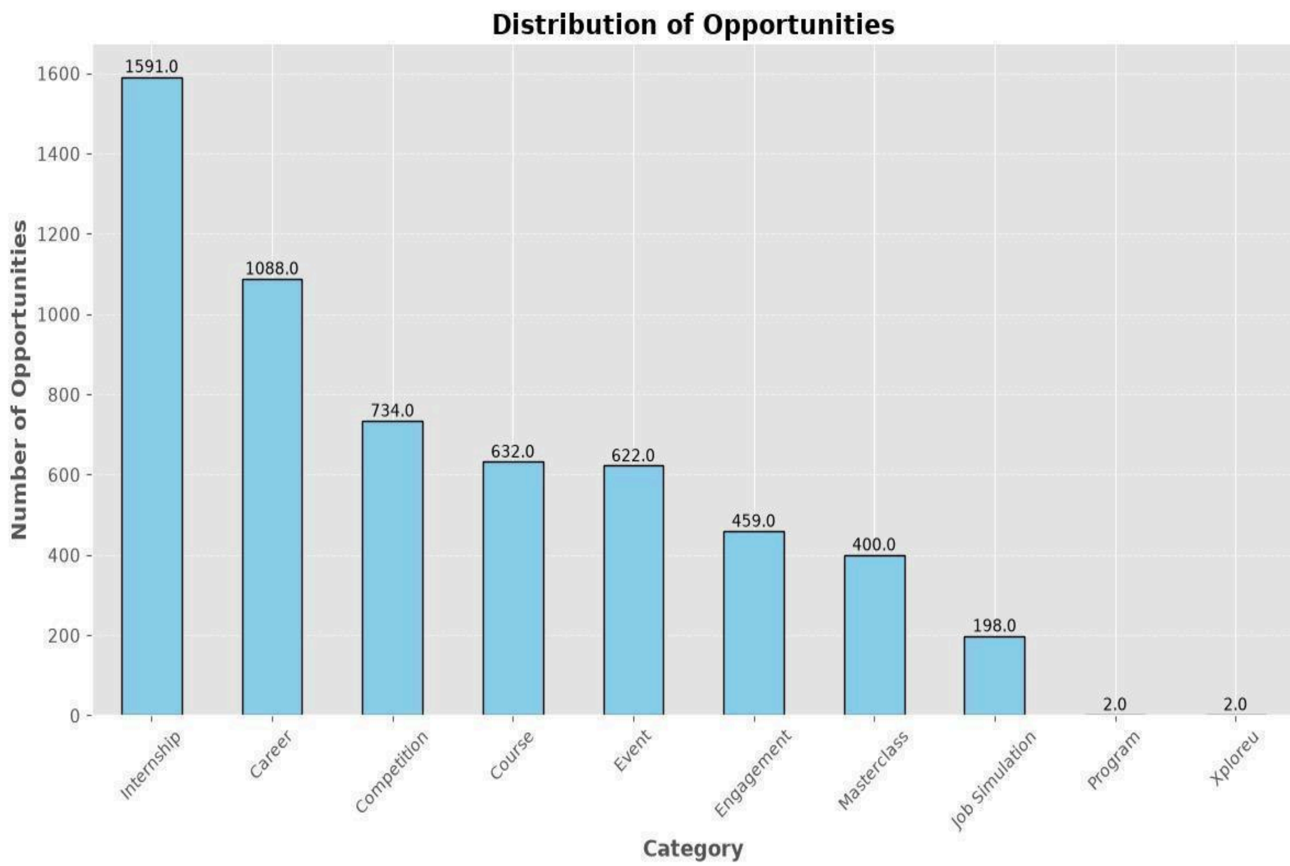

Figure 1: Presents the distribution of opportunities across the ten opportunity categories contained in the cleaned Opportunity dataset.

Explanation & Interpretation: The opportunity category distribution analysis examine the composition of opportunities available on the Excelerate platform. The analysis shows that the dataset contains 10 distinct opportunity categories, reflecting a diverse range of opportunities, including *internships, courses, competitions, engagements, career opportunities, XploreU, masterclasses, programs, job simulations, and events*. The distribution of opportunities across these categories is uneven, indicating that certain categories contribute a larger share of the dataset than others. This pattern sugg+ests that the Excelerate platform places greater emphasis on selected opportunity types, particularly those associated with employability, skills development, and professional growth. The category distribution therefore provides important context for interpreting subsequent analyses, as trends observed in the dataset may be influenced by the relative concentration of opportunities within specific categories.

Relevant Summary Statistics

```python
df.describe () # Display the statistical summary of the dataset, including mean, median, standard deviation, minimum, maximum, and quartile
```

[21] Python

	created_at	duration	fee	last_date_to_apply	microscholarship	modified_at
count	5727	5728.000000	5.728000e+03	5727	5.727000e+03	5726
mean	2024-06-17 04:46:55.854723	30.346892	1.047486e+09	2024-04-05 20:51:10.647808	8.922746e+04	2024-07-20 01:01:43.527768
min	2022-08-08 23:06:40	0.000000	0.000000e+00	1970-01-01 00:00:00	0.000000e+00	2023-03-28 10:40:00
25%	2023-11-14 22:13:20	12.000000	0.000000e+00	2023-11-14 22:13:20	1.000000e+02	2023-11-14 22:13:20
50%	2024-03-09 16:00:00	30.000000	0.000000e+00	2024-07-03 09:46:40	1.200000e+02	2024-03-09 16:00:00
75%	2025-02-19 21:20:00	30.000000	0.000000e+00	2025-02-19 21:20:00	1.200000e+02	2025-02-19 21:20:00
max	2026-05-28 20:26:40	11334.000000	6.000000e+12	2027-12-28 13:20:00	5.000000e+08	2026-05-28 20:26:40
std	NaN	215.079570	7.927746e+10	NaN	6.608328e+06	NaN

Figure 2: Detailed descriptive summary of numerical columns

Explanation & Interpretation: The summary statistics show substantial variation across the numerical variables in the Opportunity dataset. The **Duration** variable recorded a **mean of 30.35** days and a **median of 30.00 days,** indicating a relatively consistent typical duration for most opportunities. In contrast, the **Fee** and **Microscholarship** variables displayed large differences between their means and medians, reflecting the presence of extreme outlier values. The Fee variable recorded a median of **0.00** but a maximum value of **6,000,000,000,000.00,** while the Microscholarship variable recorded a median of **120.00** and a maximum value of **500,000,000.00.** These extreme values inflate the mean and make it less representative of the typical opportunity fee or scholarship amount. Consequently, the median was used as the primary measure of central tendency for Fee and Microscholarship, as it provides a more reliable representation of the typical values within the dataset.

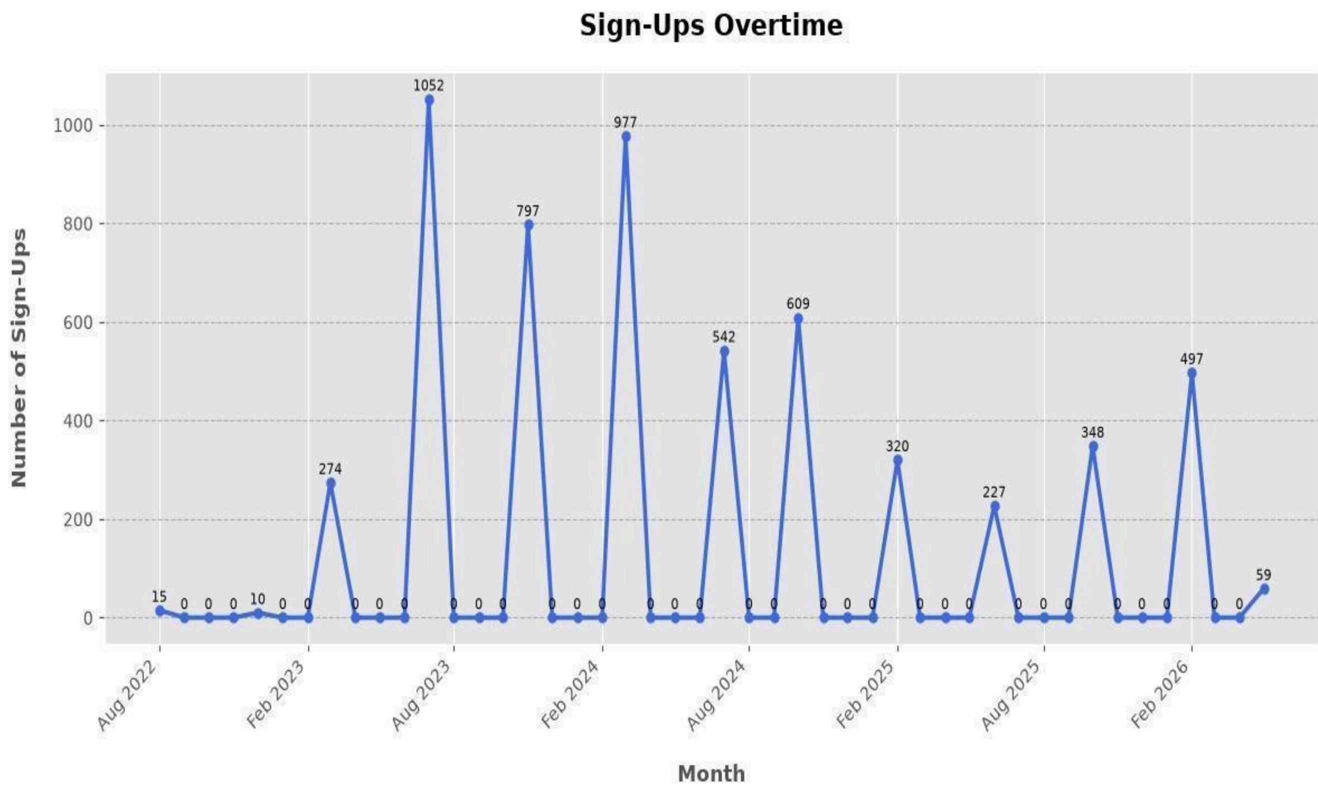

Figure 3: This figure illustrates the monthly distribution of opportunities created based on the `created_at` variable.

Explanation & Interpretation: The opportunity sign-up trend analysis was conducted using the `created_at` variable to examine the temporal distribution of opportunity publication. Monthly aggregation of the records revealed that opportunity creation was not evenly distributed throughout the study period, with distinct periods of higher and lower publication activity. The pattern reflects fluctuations in platform activity rather than a constant rate of opportunity creation. Internship opportunities constitute a substantial proportion of the opportunities represented in the dataset, indicating that internship-related publication activity contributes significantly to the overall monthly trend. The observed variation in opportunity creation therefore reflects both the concentration of internship opportunities and the broader cyclical pattern of opportunity publication across the platform.

Analysis of Applications by Location

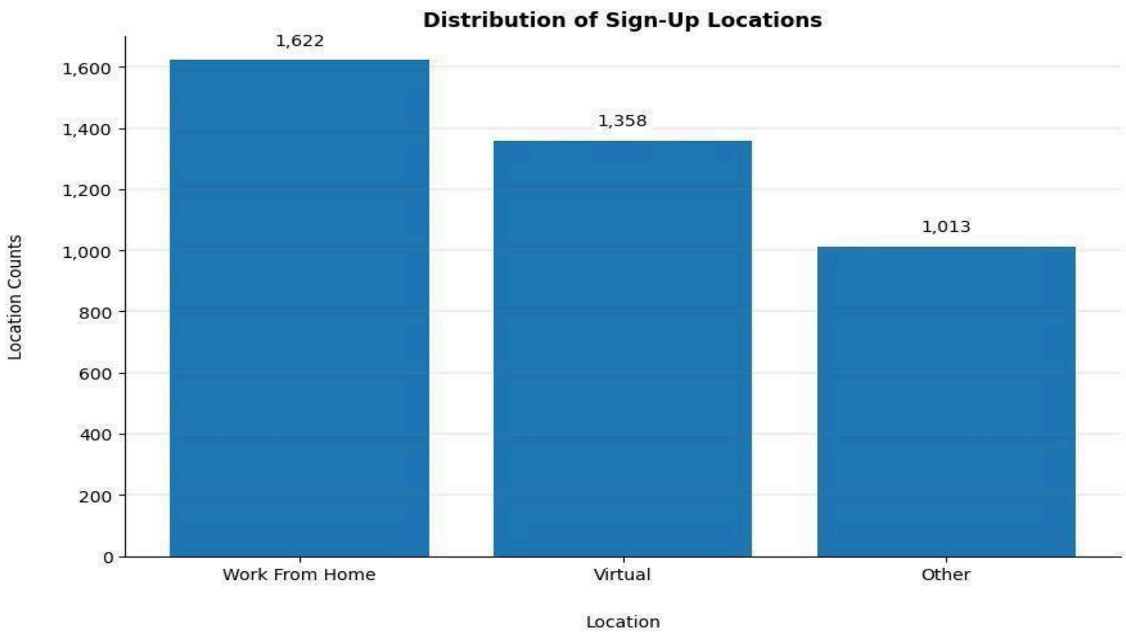

Figure 4: Presents the distribution of opportunities across the recorded location categories.

Explanation & Interpretation: The analysis of applications by location was conducted using the `location` variable to examine the distribution of opportunities across different delivery modes and geographic categories. Locations that could not be validated or had very few occurrences were grouped under Other to simplify the analysis and improve the clarity of the visual presentation. The distribution shows that **Work From Home a**nd **Virtual** opportunities dominate the dataset, with **Work From Home** recording **1,622** opportunities and **Virtual** recording **1,358** opportunities, compared to **1,013** opportunities grouped under Other.

Figure 5: shows the distribution of opportunity categories across different locations and delivery modes.

Interpretation: The heatmap further confirms that these remote delivery modes contain the highest concentration of opportunities across multiple categories, particularly within *Internship, Career, Competition, Course, Event, and Masterclass* opportunities. The concentration of opportunities in **Work From Home** and **Virtual** locations indicates that the Excelerate platform is strongly oriented toward remote and flexible opportunity delivery. This pattern reflects a broad availability of opportunities that can be accessed without physical location constraints, while the smaller number of opportunities in the **Other** category suggests limited representation of less common or unvalidated locations.

Analysis of Outreach Channel Performance

The cleaned Opportunity dataset was reviewed to identify variables related to outreach channel performance, such as email campaigns, social media platforms, referrals, or advertising sources. No such variables were present in the retained dataset. Consequently, the analysis could not evaluate the effectiveness of specific outreach channels or determine which channels contributed most to opportunity visibility and applicant engagement. This limitation indicates that outreach channel performance falls outside the scope of the current exploratory data analysis and would require additional operational or marketing data for meaningful evaluation.

Analysis of Scholarship Distribution

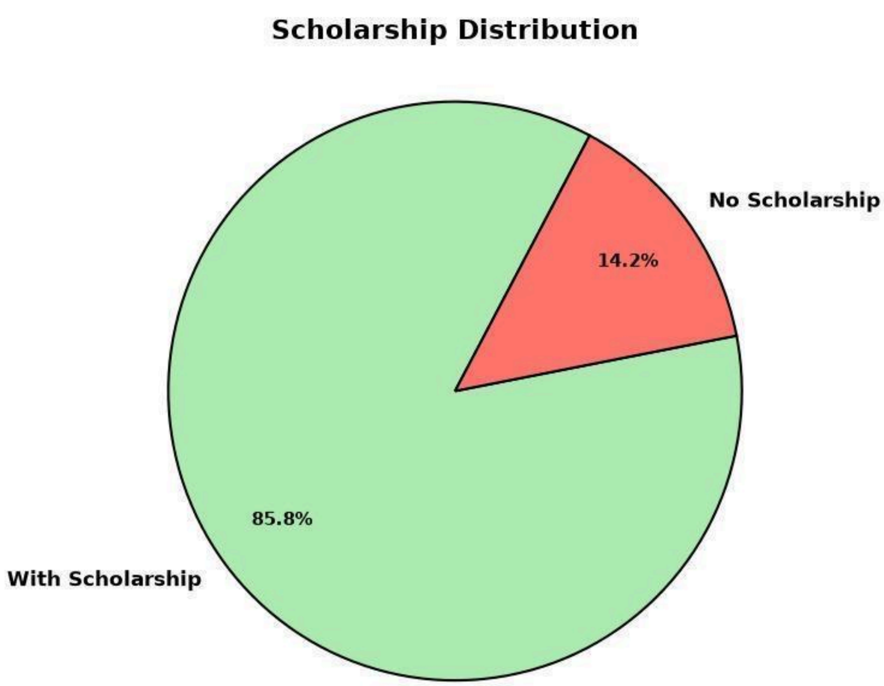

Figure 6: presents the proportion of opportunities with and without microscholarship support.

Explanation & Interpretation: The scholarship distribution analysis was conducted to examine the proportion of opportunities offering microscholarship support within the cleaned Opportunity dataset. The pie chart shows that 85.8% of opportunities provide a microscholarship, while 14.2% do not offer scholarship support. The distribution indicates that scholarship-supported opportunities form the majority of the dataset, suggesting that microscholarship provision is a common feature across opportunities available on the Excelerate platform. The relatively small proportion of opportunities without scholarship support highlights the platform's strong concentration of opportunities that include some form of financial incentive or support for participants.

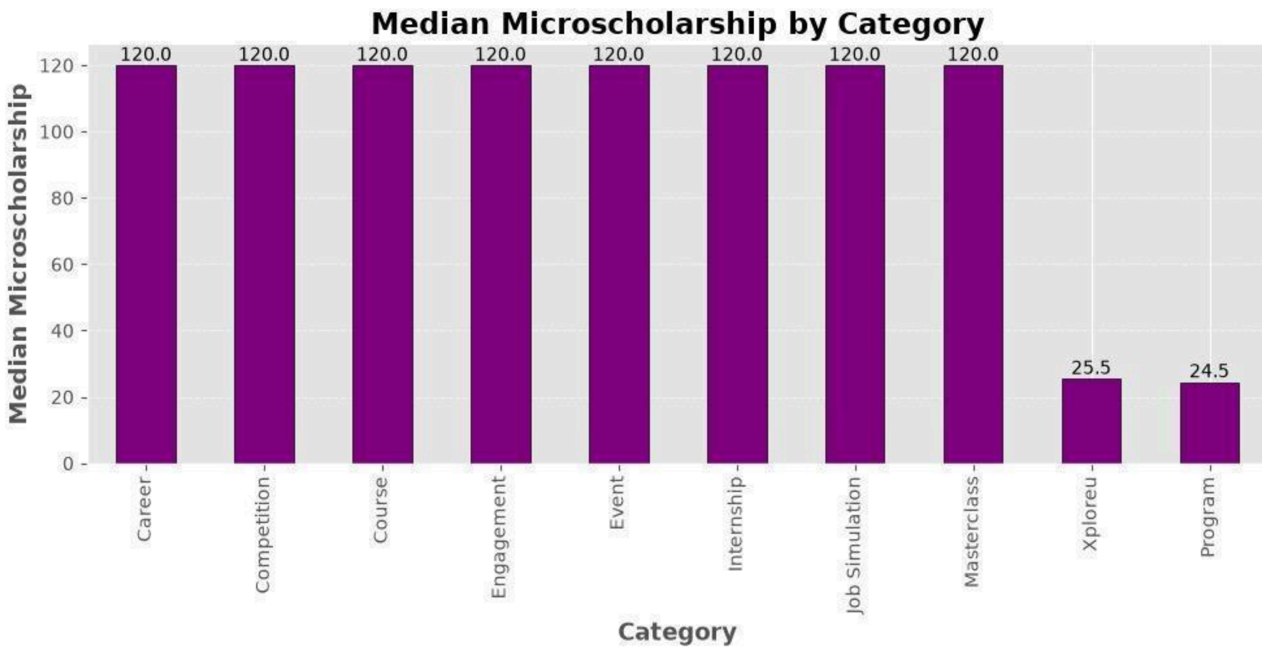

Figure 7: shows the median microscholarship amount across different opportunity categories.

Explanation & Interpretation: The median microscholarship analysis examine the typical scholarship amount offered across different opportunity categories. The analysis shows that most categories recorded a median **microscholarship** value of approximately **120**, indicating that the typical scholarship amount is broadly similar across these categories. However, Program and XploreU opportunities recorded lower median values of **24.5** and **25.5**, respectively, reflecting comparatively lower typical scholarship amounts.

Analysis of Duration by Opportunity Category

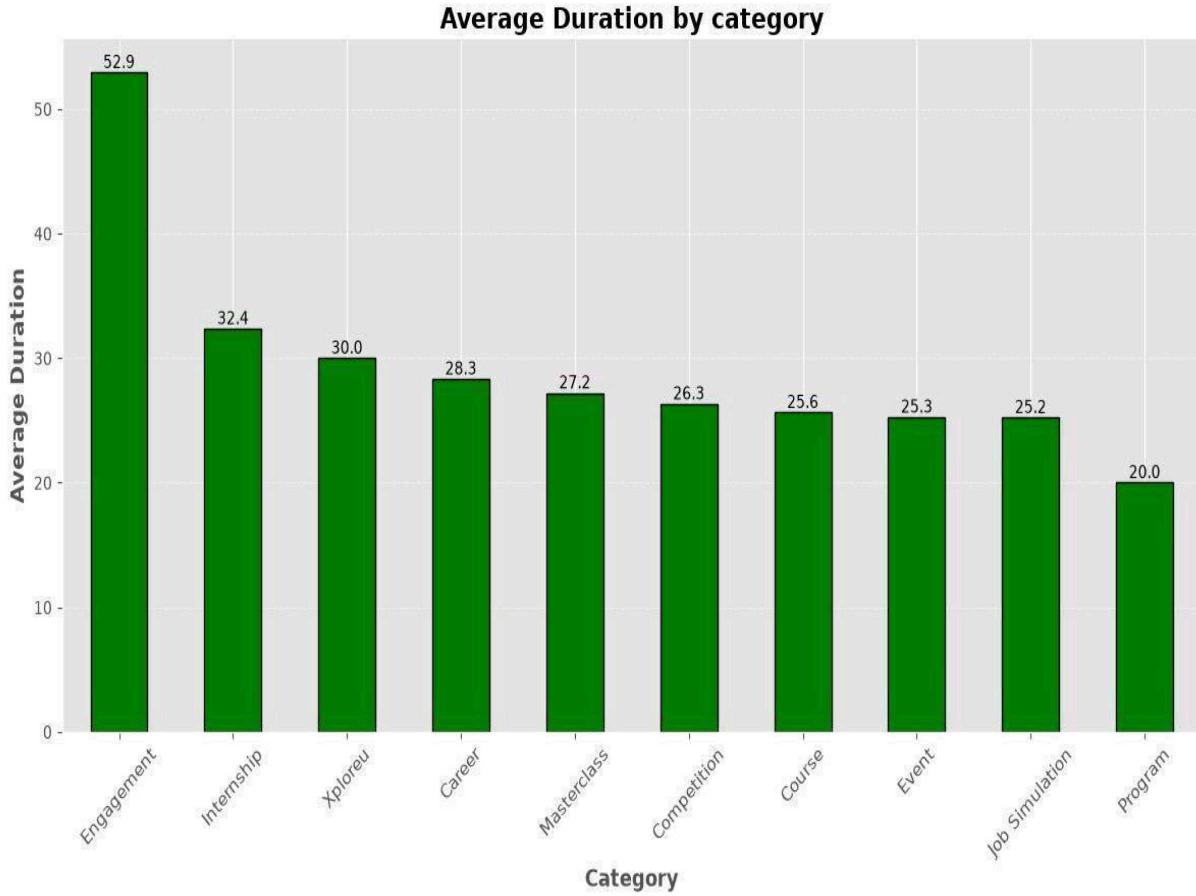

Figure 8: figure presents the average duration of opportunities across different opportunity categories.

Explanation & Interpretation: The average duration analysis reveals clear variation in the length of opportunities across categories. **Engagement** opportunities recorded the **longest** average duration, indicating that they generally remain active for a longer period than other opportunity types. In contrast, **Program** opportunities recorded the **shortest** average duration, suggesting a comparatively shorter participation or application period. These differences demonstrate that opportunity categories are structured with varying time commitments, reflecting the distinct objectives and delivery formats associated with each category.

Analysis of Auto Approval Status by Opportunity Category

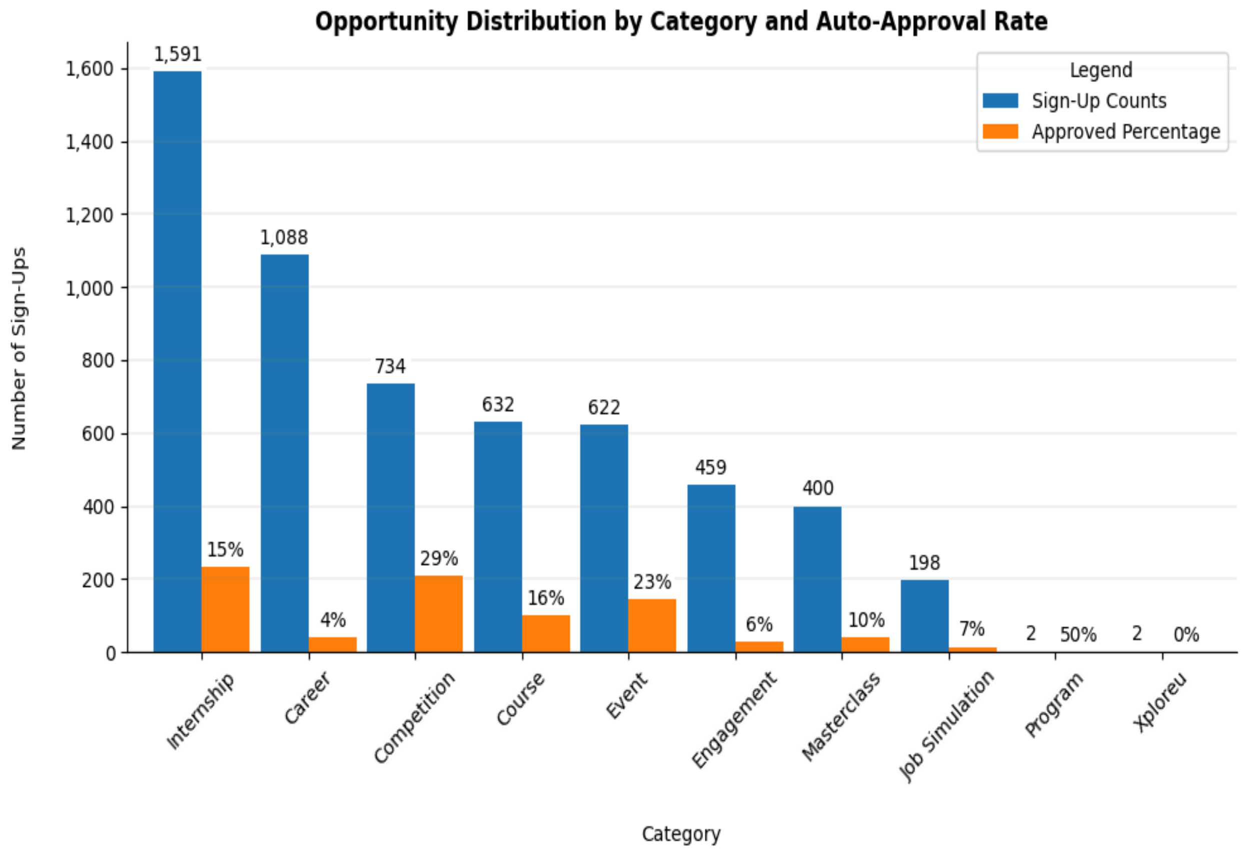

Figure 9: This figure compares the total number of sign-ups and the proportion of auto-approved opportunities across different opportunity categories.

Explanation & Interpretation: The analysis shows that **Internship** opportunities constitute the largest category in the dataset, with **1,591** opportunities, followed by **Career (1,088)** and **Competition (734)** opportunities. However, the proportion of auto-approved opportunities varies considerably across categories. **Program** opportunities recorded the highest auto-approval rate at **50%**, although this category contains only **2** opportunities, making the percentage less reliable for broad interpretation. Among the larger categories, **Competition** opportunities recorded the highest auto-approval rate at **29%**, followed by **Event** opportunities at **23%**. In contrast, **Career** opportunities recorded a relatively low auto-approval rate of **4%,** while **XploreU** recorded **0%** auto approval. Overall, the visual indicates that most opportunities are not automatically approved, suggesting that the platform predominantly relies on a review or selection process before applicants are accepted.

Analysis of Median Fee by Opportunity Distribution

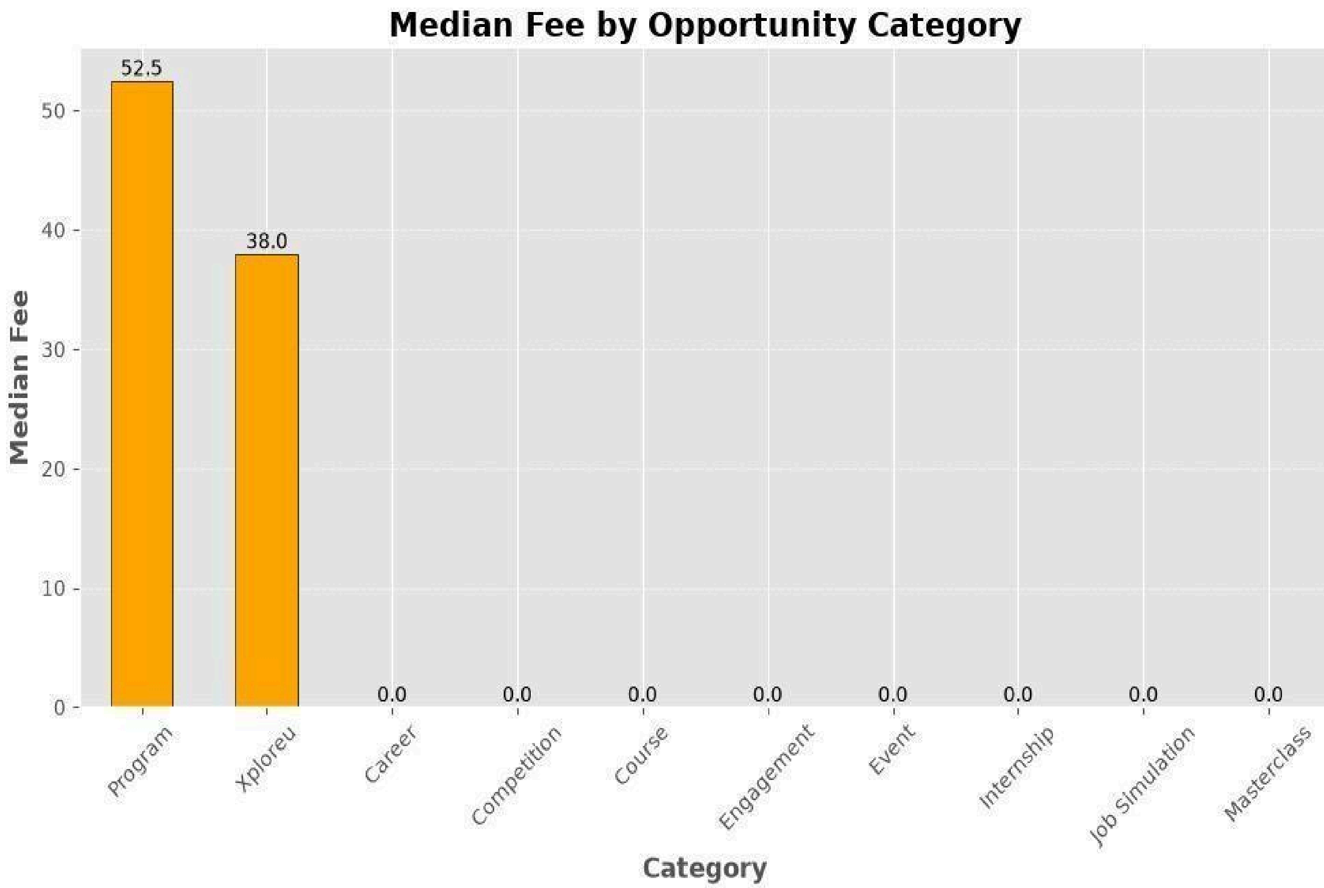

Figure 10: Presents the median application fee across different opportunity categories.

Explanation & Interpretation: The median fee analysis shows that **Program** opportunities recorded the highest typical application fee at **52.5**, followed by **XploreU** opportunities at **38.0**. In contrast, the remaining opportunity categories recorded a median fee of 0.0, indicating that at least half of the opportunities within those categories did not require an application fee. The use of the **median** rather than the **mean** provides a more reliable measure of the typical fee level, as the Fee variable contained extreme outlier values that significantly inflated the average. These findings suggest that most opportunities on the Excelerate platform are generally free to apply for, while Program and XploreU opportunities are more likely to involve a typical application cost.

Analysis of Application Deadline by Monthly Distribution

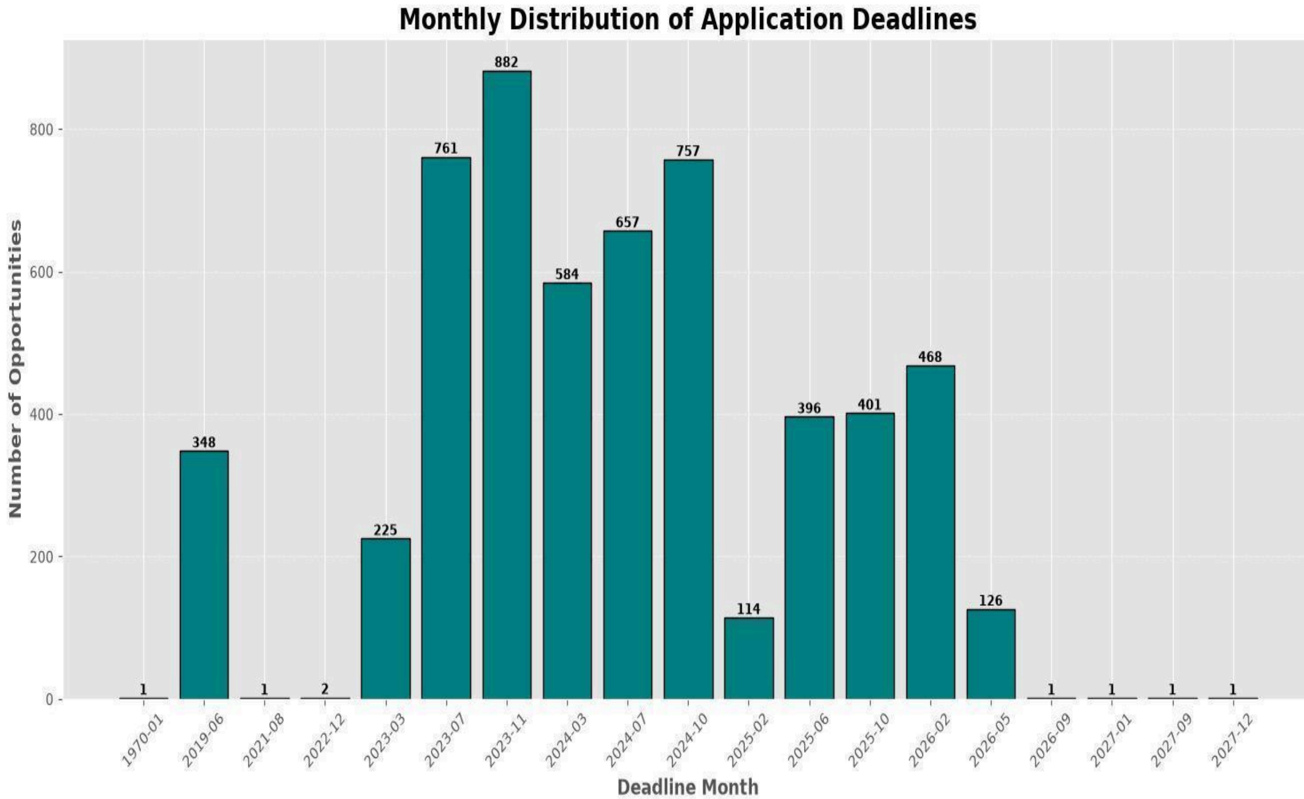

Figure 11: Presents the monthly distribution of application deadlines based on the last_date_to_apply variable.

Explanation & Interpretation: The application deadline analysis reveals that deadlines are concentrated in specific periods rather than being evenly distributed across all months. The highest number of deadlines was recorded in **November 2023 (882 opportunities)**, followed by **July 2023 (761 opportunities)** and **October 2024 (757 opportunities)**. This concentration suggests that applicants are likely to encounter periods of increased application activity, during which a larger number of opportunities close within a relatively short timeframe. The pattern also indicates that application deadlines tend to cluster around the same periods in which opportunity creation activity was high.

Analysis of Currency Types Distribution

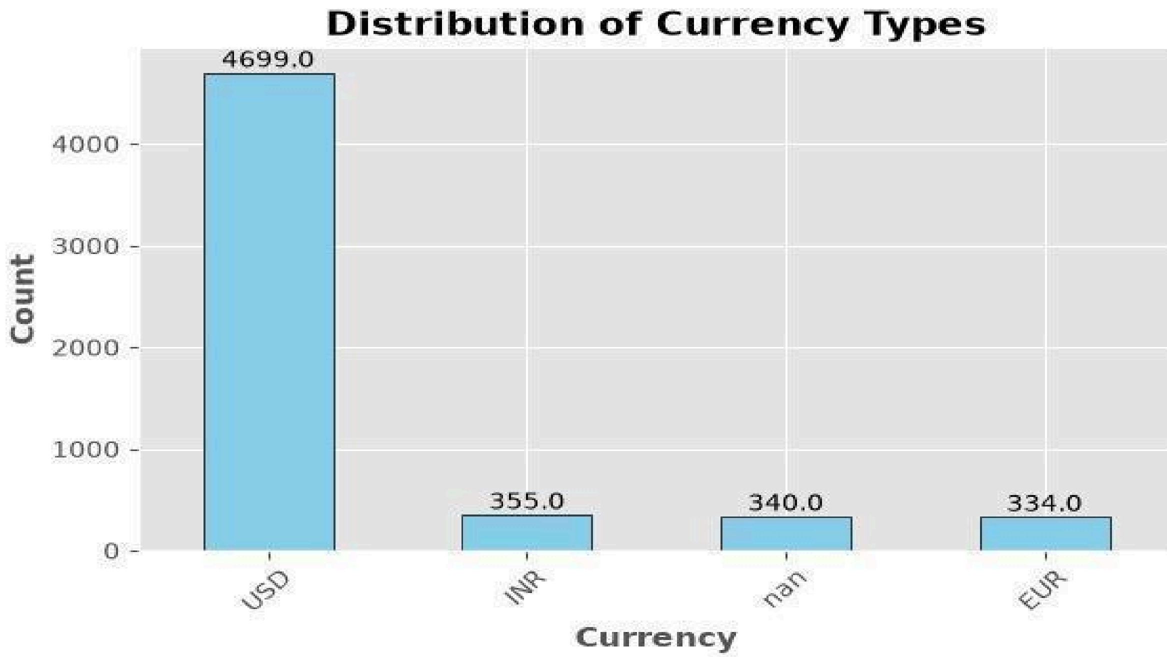

Figure 12: This figure presents the distribution of currency types used for opportunity fees in the cleaned Opportunity dataset.

Explanation & Interpretation: The currency distribution analysis shows that **USD** is the dominant currency used for opportunity fees, with **4,699** recorded opportunities, far exceeding the counts for **INR (355)** and **EUR (334).** A smaller number of records also contain missing or unclassified currency information **(about 340).** This distribution indicates that the majority of fee-based opportunities on the Excelerate platform are denominated in **USD**, suggesting a strong international orientation in the platform's fee structure.

3. Key Findings Summary

Opportunity Availability: The dataset was dominated by *Internship*, *Career*, and *Competition* opportunities, indicating that the Excelerate platform places strong emphasis on employability and professional development opportunities. Opportunity creation and application deadlines were concentrated in identifiable peak periods, rather than being evenly distributed across the study period.

Accessibility of Opportunities: The analysis showed that **Work From Home** and **Virtual** opportunities formed the largest share of location-based opportunities, demonstrating that the platform primarily provides opportunities through remote and flexible delivery modes. This broadens accessibility for applicants regardless of geographical location.

Financial Support and Fees: Most opportunities (**85.8%**) included microscholarship support, highlighting financial assistance as a common feature of the platform. However, the **Fee** and **Microscholarship** variables contained extreme outlier values that inflated the mean calculations. For this reason, the **median** was used as the primary measure of central tendency, revealing that **Program** and **XploreU** opportunities recorded the highest typical application fees, while most other categories had a median fee of **0**.

Duration and Approval Patterns: Several opportunity categories, including *Internship*, *Competition*, *Course*, *Event*, and *Job Simulation*, exhibited relatively similar average durations, suggesting comparable application or participation periods. In contrast, *Engagement* opportunities recorded the longest average duration, while *Program* opportunities recorded the shortest. The analysis also showed that most opportunities were *not automatically approved*, indicating that applicants generally undergo a review or selection process before acceptance.

Cross-Analytical Insight: A comparison of category distribution, fee, and duration revealed that **Program** and **XploreU** opportunities, which recorded the highest typical application fees, also represented a comparatively smaller share of opportunities in the dataset. In particular, **Program** opportunities combined the highest median fee with the shortest average duration, suggesting a shorter application period for opportunities that are more likely to involve an application cost.

Overall Readiness for Future Analysis: Overall, the exploratory data analysis identified meaningful patterns in opportunity availability, accessibility, financial support, duration, fees, and approval status. Despite the presence of some missing values and extreme outliers, the cleaned dataset is sufficiently structured and reliable for further exploratory analysis and future decision-making activities.

4. Data Limitations and Future Analysis

Data Completeness: Although the cleaned dataset was suitable for exploratory analysis, the relatively high proportion of missing values in the **Location** variable limits the precision of geographic and accessibility-related findings. Improving the completeness of location information would strengthen future location-based analyses.

Analytical Scope: The dataset represents **opportunities rather than individual applicants**, which restricted the analysis to opportunity-level patterns. Consequently, applicant behaviour, including application volume, completion rates, conversion rates, and demographic participation, could not be evaluated.

Missing Outreach Channel Information: The required analysis of **outreach channel performance** could not be conducted because the dataset did not contain variables related to marketing or referral channels, such as email campaigns, social media sources, or advertising platforms. Including these variables in future datasets would enable a more comprehensive evaluation of applicant acquisition strategies.

Data Quality Validation: The presence of extreme outlier values in the **Fee** and **Microscholarship** variables indicates the need for further validation of unusually large records. Confirming whether these values represent legitimate high-value opportunities or data entry anomalies would improve the reliability of future financial analyses.

Future Analytical Opportunities: Future analysis should focus on exploring interactive relationships between opportunity categories, fees, scholarships, duration, deadlines, and locations. If applicant-level and outreach-channel data become available, the analysis could be expanded to evaluate applicant engagement, conversion behaviour, and the effectiveness of different recruitment channels.

5. Summary of Analytical Assessment

This exploratory data analysis examined the cleaned Opportunity dataset (Opportunity_Data_cleaned.xlsx) to identify key patterns, trends, and relationships across opportunity categories, publication activity, application deadlines, locations, scholarships, fees, duration, approval status, and currency distribution. The analysis revealed meaningful differences in opportunity availability and characteristics, with Internship, Career, and Competition opportunities forming the largest share of the dataset, while Work From Home and Virtual opportunities demonstrated the platform's strong emphasis on accessible and flexible participation.

The analysis also highlighted the Importance of using appropriate statistical measures when interpreting financial variables. Extreme outlier values in the Fee and Microscholarship variables significantly affected the mean calculations; therefore, the median was used as a more reliable measure of the typical values within the dataset. In addition, the findings showed that most opportunities include microscholarship support and that the majority of opportunities are not automatically approved, indicating the prevalence of review or selection processes.

Despite the presence of some missing values and extreme outliers, the cleaned dataset is sufficiently structured and reliable for exploratory analysis. The findings generated through this report provide a strong analytical foundation for dashboard development and future investigations into opportunity accessibility, applicant engagement, and platform performance.

6. Recommendations

Based on the findings above, three actions are recommended. First, given that 85.8% of opportunities include microscholarship support and most categories carry no application fee, marketing and outreach materials should foreground "low-cost" or "no-cost to apply" messaging to support acquisition. Second, because Program and XploreU opportunities combine the highest fees with the shortest durations, these categories should be flagged for a targeted review of pricing and conversion rates, since they diverge structurally from the rest of the platform. Third, the 25.12% missing-location rate should be addressed at the point of data entry — e.g. making location a required field — before the next reporting cycle, since it currently limits the reliability of any accessibility-focused analysis.